

EBU6505 – Reasoning and Agents

Block 1, Day 3 – Knowledge Graph Foundations for AI

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Teaching Week 3

What have we covered so far this week?

- MDPs describe sequential decision problems.
- Sequential decision models use reward to compare future choices.
- Knowledge representation stores facts in a machine-usable form.
- Triples, identifiers, and schema help keep those facts clear.

Main idea: Day 1 was about acting over time. Day 2 was about storing facts clearly. Today we connect many facts into one graph.

What are we going to cover today?

- How single triples become connected knowledge graphs
- How paths in a graph help answer questions
- Why IDs, provenance, freshness, and quality checks matter
- How modern AI systems retrieve small grounded pieces of a graph

Main idea: The goal is simple: build a graph that an AI system can search, trust, and explain.

- *Artificial Intelligence: Foundations of Computational Agents*, Poole and Mackworth (3rd ed.)
- Chapters 16 and 17 – graph structure, knowledge graphs, and reasoning over linked facts
- *Artificial Intelligence: A Modern Approach*, Russell and Norvig (4th ed.)
- Chapter 10 – knowledge representation

Main idea: The books give the formal view. In class, we focus on small examples that show how graph reasoning works in practice.

Learning Outcomes for Today

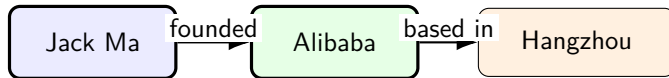
By the end of this session, you should be able to:

- explain how a knowledge graph differs from one isolated triple
- describe how graph paths support question answering and retrieval
- explain why identifiers, provenance, and freshness are important
- sketch a basic workflow for building and maintaining a small knowledge graph

Warm-up Exercise

Question: If the graph knows that Jack Ma founded Alibaba and Alibaba is in Hangzhou, what new question can it answer?

- Read the two connected facts.
- Look for a path through the graph.
- Say the answer in one short sentence.



Main idea: A graph becomes useful when two simple facts connect and support a new question.

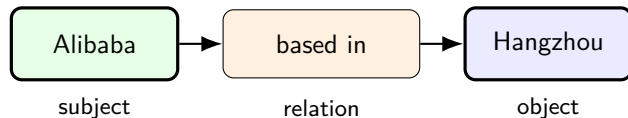
From One Fact to One Triple

Question: How do we turn one sentence into a graph-friendly fact?

Sentence

Alibaba is based in Hangzhou.

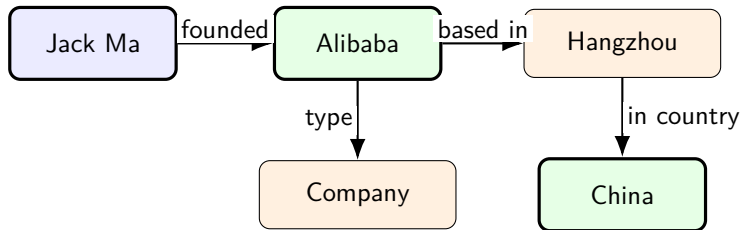
- Who or what is the main entity?
- What relation is being stated?
- What is the linked value or entity?



Main idea: A triple keeps one fact short, explicit, and easy for a machine to reuse later.

From Many Triples to One Graph

Question: What changes when several triples share the same entities?



Main idea: A knowledge graph is not just one fact. It is a network of facts that share entities and relations.

Entities, Relations, and Attributes

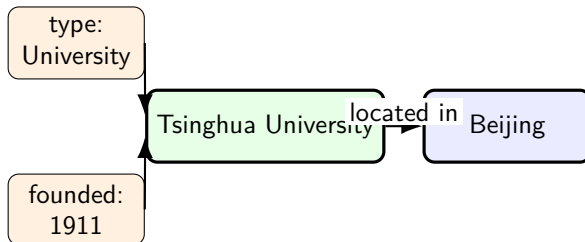
Question: What kinds of things do we usually store in a knowledge graph?

Three core parts

Entity: main thing

Relation: named link

Attribute: property value



Main idea: Entities are the main nodes. Relations connect entities. Attributes add small details.

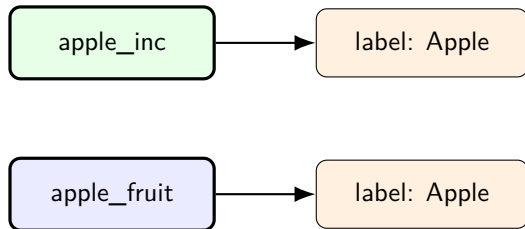
Why IDs Matter

Question: Why is the name alone often not enough for a graph?

Name only

“Apple” can mean:

- the company
- the fruit



Main idea: A good graph gives each entity a stable ID, even when two entities share the same name.

Relation Design Must Be Precise

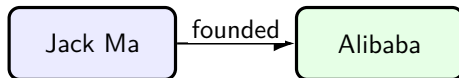
Question: Why are vague relation names bad for retrieval and reasoning?

Too vague

- related to
- connected to
- associated with

Clearer

- founded
- based in
- taught by



Main idea: If relation names are clear, the system can retrieve the right path and explain the answer more easily.

Pattern Versus Real Facts

Question: What is the difference between a graph pattern and one actual set of facts?

Schema idea

Person – founded – Company – based
in – City

Instance idea

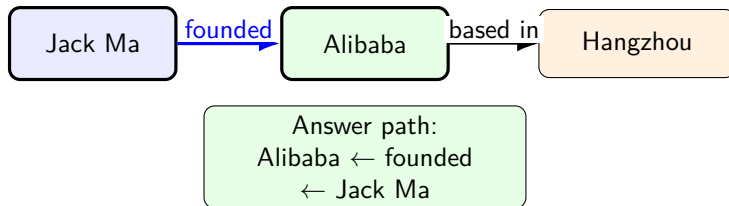
Jack Ma – founded – Alibaba – based
in – Hangzhou



Main idea: The schema gives the pattern. The instance graph fills that pattern with real entities and real facts.

A Graph Path Can Be the Answer

Question: How can the graph answer the question “Who founded Alibaba?”



Main idea: The system can follow the right graph edge instead of scanning unrelated text.

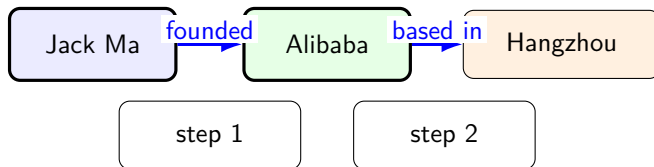
Multi-Hop Reasoning

Question: How can two connected edges answer a harder question?

Question

Which city is the company founded by Jack Ma based in?

- Step 1: find the company founded by Jack Ma.
- Step 2: follow the next edge from that company.



Main idea: Multi-hop reasoning means following more than one edge to reach the answer.

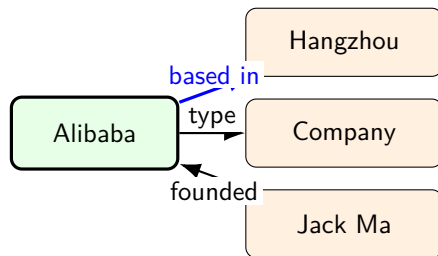
Neighborhood Retrieval for AI

Question: When a user asks one question, should the system read the whole graph?

User query

Where is Alibaba based?

- Link the query to the entity Alibaba.
- Retrieve only the nearby facts.
- Highlight the edge that answers the question.



Main idea: Good retrieval starts from the right entity and keeps only the useful neighborhood.

Common Data-Quality Problems

Question: What can go wrong if a graph is badly maintained?

Wrong fact

Example: wrong city stored

Duplicate entity

Example: one person stored twice

Missing link

Example: headquarters link missing

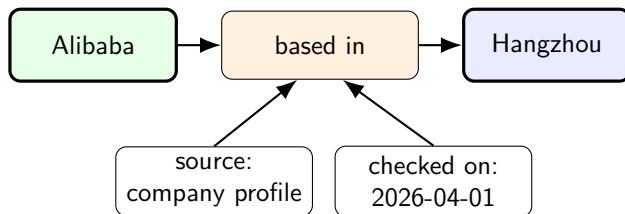
Old fact

Example: location changed, graph not updated

Main idea: A graph is useful only when its facts are correct, complete enough, and up to date.

Why Provenance Matters

Question: How do we know where one graph fact came from?



Main idea: Provenance tells us the source and date of a fact, so we can trust it, check it, and explain it.

Freshness Matters Too

Question: Why is a correct old fact still a problem?

The diagram consists of two rounded rectangular boxes. The left box is pink and contains the text: 'Older record', 'office location', '= Room 101', and 'checked in 2024'. The right box is green and contains the text: 'Current record', 'office location', '= Room 204', and 'checked in 2026'. A black arrow points from the pink box to the green box, with the text 'update over time' written above it.

Record Type	Office Location	Checked In
Older record	Room 101	2024
Current record	Room 204	2026

Main idea: A graph should not just store facts. It should also know when those facts were last checked.

Entity Resolution

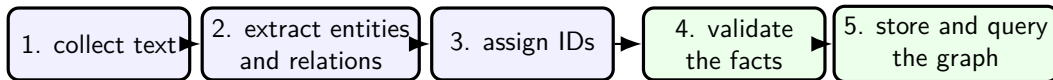
Question: How do we avoid storing the same real entity more than once?



Main idea: Entity resolution is the process of deciding whether two records refer to the same real thing.

How to Build a Small Practical KG

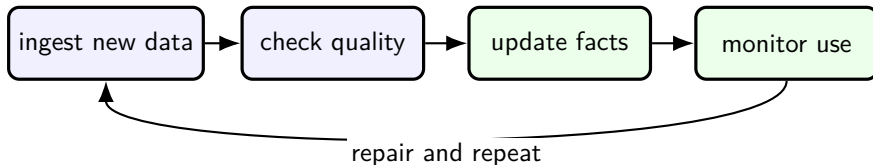
Question: What are the main steps from raw text to a usable graph?



Main idea: Building a graph is a pipeline: extract, identify, validate, then store.

A Graph Needs Maintenance, Not Just Ingestion

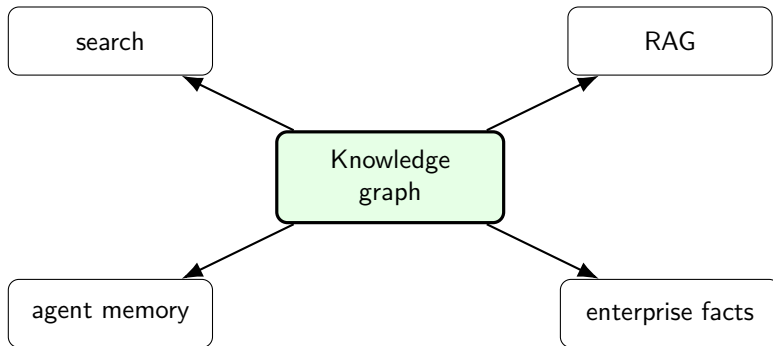
Question: What happens after the first version of the graph is built?



Main idea: A real knowledge graph is a living system. It needs checking, updating, and repairing over time.

Why Knowledge Graphs Matter in Modern AI

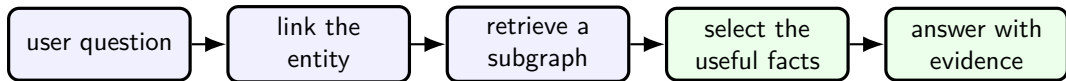
Question: Why are knowledge graphs still useful when we already have large language models?



Main idea: LLMs are strong language models. Graphs give them explicit and grounded facts.

KG-to-Agent Context Pipeline

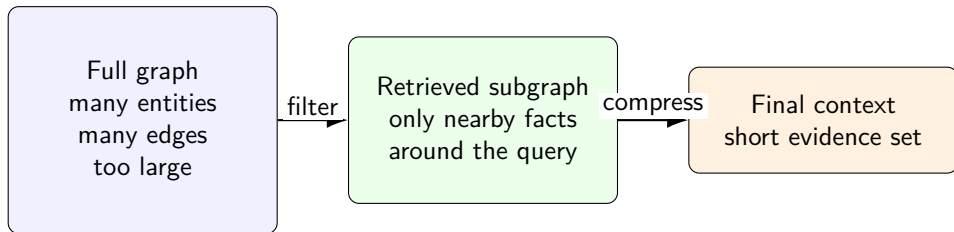
Question: How does an agent turn a user question into grounded context from a graph?



Main idea: The agent should not answer first and search later. It should retrieve evidence first, then answer.

Do Not Send the Whole Graph

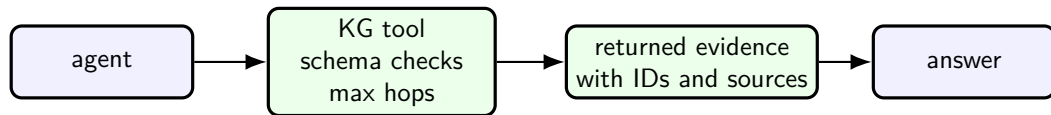
Question: Why do we need to reduce the graph before giving context to an LLM?



Main idea: A good system narrows the graph down to only the evidence that fits the question and the context budget.

Grounded Tool Use Needs Guardrails

Question: How do we stop an agent from inventing graph facts that were never retrieved?

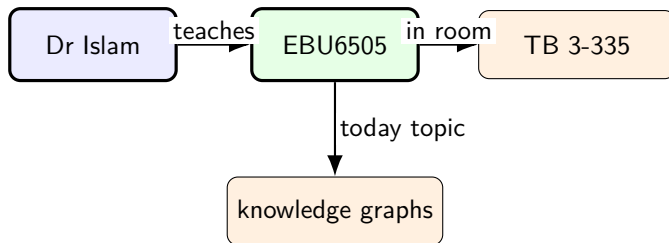


Require the answer to stay close to the retrieved graph evidence.

Main idea: Guardrails keep the agent grounded in retrieved facts instead of imagined facts.

Campus Assistant Example

Question: What could a small teaching assistant graph look like?



- A student can ask: Who teaches EBU6505?
- A student can ask: Where is the class?
- A student can ask: What topic is today?

Main idea: Even a small graph can support several useful questions when the entities and relations are clear.

Mini Design Exercise

Question: If you built a graph for a campus assistant, which entities and relations would you include first?

Possible entities

module, lecturer, room, student, deadline

Possible relations

teaches, in room, has deadline, enrolled in

Task

Write:

- 4 entity types
- 4 relation types
- 2 example questions your graph should answer

Main idea: Design starts from the questions you want the graph to answer.

Transition to Day 4

Question: Once the graph exists, how do search, RAG, and modern agents use it at run time?

- Today focused on the structure and maintenance of the graph.
- The next lecture focuses on using that graph inside modern AI pipelines.
- We will move from graph foundations to graph use in retrieval and reasoning.

Main idea: Day 3 is about building the graph well. Day 4 is about using the graph well.

Key Takeaways

- A knowledge graph is a connected set of machine-usable facts.
- Graph paths support one-hop and multi-hop question answering.
- IDs, provenance, freshness, and entity resolution are essential, not optional.
- Modern AI systems use graphs as grounded evidence stores for retrieval and reasoning.

Main idea: If the graph is clear and well maintained, it becomes a strong foundation for knowledge-based reasoning and RAG.

Further Study

- Read Poole Chapter 16 and revisit AIMA Chapter 10.
- Practice: Poole Ch.16, Exercises 16.2, 16.3, 16.4, 16.5, and 16.9.
- AIMA knowledge-representation exercise page: 3, 7, 8, 16, and 26.
- Focus on schema versus instance facts, connected triples, and what provenance adds to an answer.